Task 1:

1. Introduction to Radiance Values

What is Spectral Radiance?

Spectral radiance (L_{λ}) is the amount of energy emitted, reflected, or transmitted by a surface per unit area, per unit solid angle, and per unit wavelength. It is measured in **Watts per square meter per steradian per micrometer (W/m²/sr/µm).**

Why is Radiance Important?

Radiance values allow us to:

- Compare satellite imagery across different time periods and locations.
- Correct for sensor-specific variations by converting DN to physical energy measurements.
- Use remote sensing data for scientific analysis, including land cover classification and environmental monitoring.

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2. Converting Digital Numbers (DN) to Radiance

Understanding DN Values

Landsat satellite images are stored in **Digital Numbers (DNs),** which range from **0 to 255** for an 8-bit image. These values are **unitless** and need to be converted to radiance using calibration coefficients.

Radiance Conversion Formula

To convert DN values to radiance, we use the equation:

$$L\lambda = ML \times DN + AL$$

Where:

- Lλ = Spectral Radiance (W/m²/sr/µm)
- ML= Radiance Multiplicative Scaling Factor (Gain)
- AL= Radiance Additive Scaling Factor (Bias)

- DN = Digital Number (0-255)

The values of and depend on the Landsat sensor. For **Landsat 7 ETM+**, standard values are:

Band	Gain (/M_L/)	Bias (/A_L/)
Band 1	0.77874	-6.98
Band 2	0.79880	-7.20
Band 3	0.62165	-5.62
Band 4	0.63976	-5.78
Band 5	0.12622	-1.16
Band 6 (Thermal)	0.06709	0.00
Band 7	0.04389	-0.39

Use these values to convert DN values of Lanier image to radiance.
hint: Apply the formula manually for each band, Once all bands are processed, use **Layer Stack** to combine them into a single multispectral image.

For extra marks, students who can find a way to apply the formula **simultaneously** to all bands using **Model Maker** or **Spatial Modeler** will receive bonus points.

Task 2:

Radiometric Enhancement

Radiometric enhancements are techniques that improve contrast between certain features by altering the screen colors assigned to specific ranges of pixel values. For example, if the values of two adjacent pixels are 53 and 57 on a gray scale, and the gray scale varies linearly from 0 (black) to 255 (white), then it will impossible to visually distinguish the pixels on the screen. Both pixels will be an almost identical dark gray.

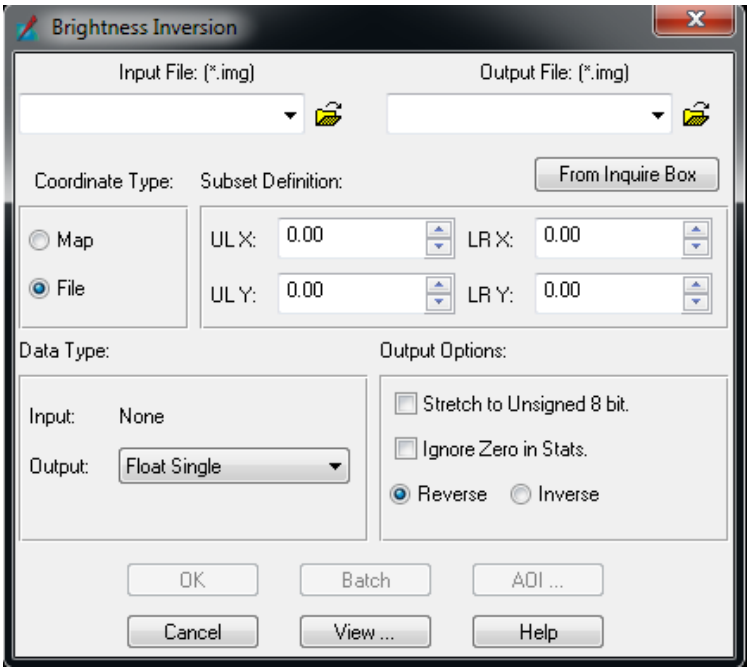
But if we change the color scale and state that all values from 0 to 50 are black, and all values from 60 to 255 are white, then the values from 51 to 59 can be "stretched" out to display a wider range of grays. In this case, the pixel with a value of 53 will be a medium-dark gray, and the pixel with a value of 57 will be a lighter gray. But this enhanced contrast in the range of 51 to 59 comes at a cost; now we cannot distinguish among the pixels with values of 0 to 50 (which are all black) or the pixels with values of 60 to 255 (which are all white) This is always the case with radiometric enhancements: the gain of contrast in one range leads to a decrease in another.

Apply Radiometric Enhancement

1. In the **Raster** tab, click **Radiometric dropdown arrow**. The **Radiometric Enhancement** dropdown opens.

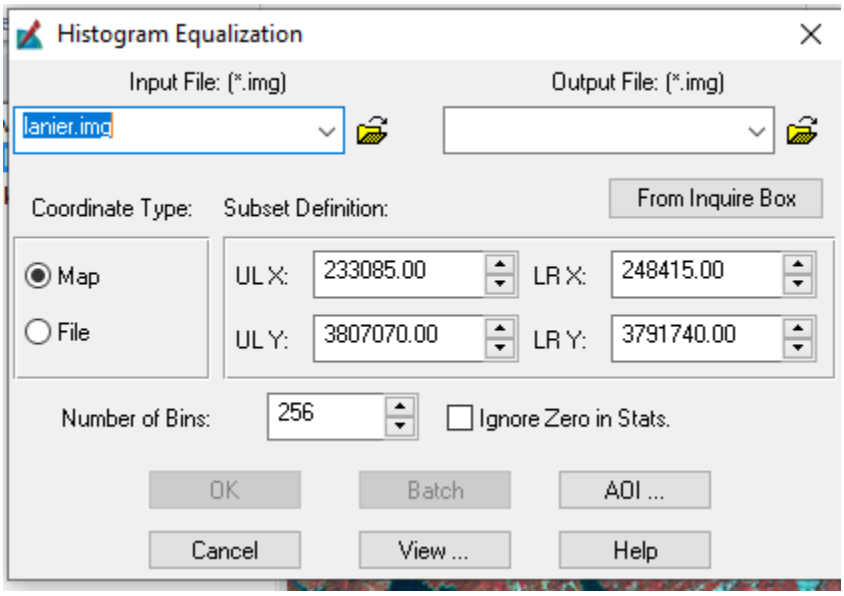
   In the brightness inversion dialog, you use both the Inverse and Reverse options of the Image Inversion function to enhance images. Inverse emphasizes detail in the dark portions of an image. Reverse simply reverses the DN values.

2. In the **Radiometric Enhancement** dropdown, select **Brightness Inversion**. The **Brightness Inversion** dialog opens.



3. In the **Brightness Inversion** dialog under **Input File**, enter **Islamabad.img**.
4. Under **Output File**, enter **inverse.img** in the directory of your choice.
5. Under **Output Options**, turn on the **Stretch to Unsigned 8 bit** checkbox by clicking on it.
6. Under **Output Options**, click **Inverse**.
7. Click **OK** in the **Brightness Inversion** dialog to start the process. A **Job Status** dialog displays, indicating the progress of the function.
8. Select **Brightness Inversion** from the **Radiometric dropdown** menu. The **Brightness Inversion** dialog opens.
9. In the **Brightness Inversion** dialog, enter **Islamabad.img** as the **Input File**.
10. Enter **reverse.img** in the directory of your choice as the **Output File**.
11. Turn on the **Stretch to Unsigned 8 bit** checkbox under **Output Options**.
12. Click **OK** in the **Brightness Inversion** dialog to start the process. A **Job Status** dialog displays, indicating the progress of the function.
13. Open a **Viewer** and display **inverse.img**.
14. Open the file **reverse.img** in another viewer.
15. Open the file **Islamabad.img**, in third viewer.
16. Note the differences between **reverse.img** and **inverse.img** compared to the original file. Use Histogram to detect difference paste snap shots of Islamabad, inversion and reverse.img histogram here.

Apply Histogram Equalization

1. In the **Raster** tab, click **Radiometric dropdown arrow**. The **Radiometric Enhancement** dropdown opens.
2. In the **Radiometric Enhancement** dropdown, select **Brightness Inversion**. The **Brightness Inversion** dialog opens.
3. 

4. Enter **Islamabad.img** under input file, give output file name of your own choice and set number of bins to 256. Click OK.